Varun Belagali

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Education

Stony Brook University, NY, USA **Aug 2022 – Present**
PhD, Computer Science (GPA 4/4). Advised by Dimitris Samaras in Computer Vision Lab.
Co-advised by Maria Vakalopoulou, Stergios Christodoulidis, Pierre Marza at CentraleSupélec, Paris.
Research Interest: **Visual representation learning with self-supervision**

R V College of Engineering, Bangalore, India **Aug 2016 – Aug 2020**
BE, Computer Science and Engineering (GPA 9.22/10).

Research Papers (selected)

1. **TICON: A Slide-Level Tile Contextualizer for Histopathology Representation Learning** [Link] arXiv 2025
Varun Belagali*, S. Kapse*, P. Marza, .. , S. Christodoulidis, M. Vakalopoulou, D. Samaras. **CVPR 2026** (Findings)
2. **CDG-MAE: Cross-view Masked Modeling using Diffusion Generated Views** [Link] arXiv 2025
Varun Belagali, P. Marza, S. Yellapragada, .. , S. Christodoulidis, M. Vakalopoulou, D. Samaras. **TMLR 2026**
3. **Gen-SIS: Generative Self-augmentation Improves Self-supervised Learning** [Link] arXiv 2024
Varun Belagali*, S. Yellapragada*, A. Graikos, .. , D. Samaras. **CVPRW 2026 -** DataCV Proceedings (Best Paper)
4. **HyperMAE: Modulating Implicit Neural Representations for MAE Training** [Link] 2023
Varun Belagali, L. Zhou, X. Li,, D. Samaras. **NeurIPS Workshop -** Self-supervised Learning
5. **Two step convolutional neural network for automatic glottis segmentation in stroboscopic videos** [Link] 2020
Varun Belagali, A. Rao, P. Gopikishore, R. Krishnamurthy, P. Ghosh. **Biomedical Optics Express**

Experience

Graduate Researcher | Computer Vision Lab, Stony Brook University, NY, USA Sep 2022 – Present
- TICON: Omni-feature masked modeling for pretraining slide encoder. Multi-teacher distillation in embedding space.
- Self-supervised learning with synthetic data. Gen-SIS: view-invariant SSL with Dino, I-BOT, SimCLR.
CDG-MAE: cross-view SSL with Masked Autoencoders.
- HyperMAE: Implicit neural representation based decoder architecture to efficiently train Masked Autoencoders.

Research Intern | Sophont, NY, USA May 2026 – Aug 2026
Pretraining looped transformers for gigapixel histopathology images. Hosted by Tanishq Mathew Abraham.

Research Intern | Dolby Laboratories, Sunnyvale, USA June 2025 – Aug 2025
Post-training Video-JEPA models for egocentric action anticipation. Hosted by Anshul Rai.

Research Associate | Indian Institute of Science, Bangalore, India Oct 2021 – July 2022
Deep Learning for glottis segmentation in stroboscopic videos. Advised by Prof. Prasanta Kumar Ghosh.

Software Engineer | Citrix, Bangalore, India July 2020 – Sep 2021
Cloud Engineering: Development of traffic manager software to handle regional outages. Lead of cloud cost optimization.

Technical Skills

Technologies: Computer Vision, Deep Learning, Self-Supervised learning, Large-scale Distributed Pre-training.
Languages & Frameworks: Python, PyTorch, Latex
Cloud Engineering: Azure, Jenkins, Splunk, New Relic

Research Talks

Self-supervised Learning with Synthetic Data. **GenBio AI**, Palo Alto, USA (Sep 2025)

Learning Correspondences from Diffusion Generated Views. **DATAIA Talk**, CentraleSupélec, Paris, France (May 2025)

Generative Self-augmentation Improves Self-supervised Learning. **MICS Lab**, CentraleSupélec, Paris, France (March 2025); **IMAGINE Lab**, École des Ponts, Paris, France (April 2025)

Fellowships

SBU CS Chairman Fellowship (2024 - 2026), DATAIA Fellowship (2025)